Course outline

1. Linear ordinary differential equations (ODEs) and initial value problems (IVPs)
2. Power series solutions to ODEs
3. Laplace transforms
4. Fourier series
5. Boundary value problems (BVPs) and linear partial differential equations (PDEs)
6. Fourier transforms

Topic 3: Laplace transforms

How Topic 3 builds on Topic 2:

- Functions in, and solutions to, the governing ODEs in Topics 1 and 2 are generally continuous – or at least, we focus on intervals where these functions are continuous.
- In Topic 3, we tackle a new class of ODEs where discontinuous and impulsive functions are involved.

Topic 3: Laplace transforms

Leaning outcomes:

1. Understand how ODEs involving discontinuous and impulsive functions can arise in modelling of real-world problems
2. Know how to calculate Laplace transforms and inverse Laplace transforms of appropriate functions
3. Know when and how to use the Laplace transform method for solving ODEs and IVPs

Lecture 10

We will:

1. Revise definition and calculation of improper integrals
2. Learn to use comparison test to determine convergence of an integral
3. Learn the definition of the Laplace transform and its basic properties
4. Discuss application of the Laplace transform to piecewise continuous functions

Improper integrals

$$\int_a^b f(t)\mathrm{d}t$$

The above definite integral is *improper* if:

1. (a, b) is an infinite interval;
2. or f is not bounded on (a, b).

- Assume all unboundedness occurs at a and/or b (otherwise split up the integral).
- The improper integral is then defined as

$$\int_a^b f(t)\mathrm{d}t = \lim_{q\to a^+} \lim_{r\to b^-} \int_q^r f(t)\mathrm{d}t$$

Exercise 3.1: Evaluate the following improper integrals

$$\int_0^\infty \mathrm{e}^{-t}\,\mathrm{d}t \quad \text{and} \quad \int_{-1}^1 \frac{1}{t^2}\,\mathrm{d}t$$

For the first integral

$$\int_0^\infty \mathrm{e}^{-t}\mathrm{d}t = \lim_{r\to\infty}\int_0^r \mathrm{e}^{-t}\mathrm{d}t = \lim_{r\to\infty}\left[-\mathrm{e}^{-t}\right]_0^r = -\lim_{r\to\infty}\{\mathrm{e}^{-r}-1\} = 1$$

For the second integral

$$\int_{-1}^1 \frac{1}{t^2}\mathrm{d}t = \int_{-1}^0 \frac{1}{t^2}\mathrm{d}t + \int_0^1 \frac{1}{t^2}\mathrm{d}t = \lim_{r\to 0^-}\int_{-1}^r \frac{1}{t^2}\mathrm{d}t + \lim_{q\to 0^+}\int_q^1 \frac{1}{t^2}\mathrm{d}t$$

$$= \lim_{r\to 0^-}\left[-\frac{1}{t}\right]_{-1}^r + \lim_{q\to 0^+}\left[-\frac{1}{t}\right]_q^1 \ldots \text{does not exist}$$

Comparison test

Definition (Comparison test)

Let f and g be two continuous functions on (a, b).

1. If $|f(t)| \leqslant g(t)$ for all $t \in (a, b)$ then $\displaystyle\int_a^b g(t)\,\mathrm{d}t$ converges $\Rightarrow \displaystyle\int_a^b f(t)\,\mathrm{d}t$ converges
2. If $f(t) \geqslant g(t) \geqslant 0$ for all $t \in (a, b)$ then $\displaystyle\int_a^b g(t)\,\mathrm{d}t$ diverges $\Rightarrow \displaystyle\int_a^b f(t)\,\mathrm{d}t$ diverges

Exercise 3.2: Does $\displaystyle\int_0^\infty \{\sin(t)\,\mathrm{e}^{-t}\}\,\mathrm{d}t$ converge?

$$\left.\begin{array}{l} |\sin(t)\,\mathrm{e}^{-t}| \leqslant \mathrm{e}^{-t} \quad \text{for} \quad t \in (0, \infty) \\ \displaystyle\int_0^\infty \mathrm{e}^{-t}\,\mathrm{d}t \quad \text{converges (Ex 3.1a)} \end{array}\right\} \quad \int_0^\infty \{\sin(t)\,\mathrm{e}^{-t}\}\,\mathrm{d}t \quad \text{converges}$$

The Laplace transform

Definition (Laplace transform)

The *Laplace transform*, $\mathscr{L}$, of the function $f(t)$ is

$$\mathscr{L}\{f\} = \int_0^\infty \mathrm{e}^{-s\,t} f(t)\ \mathrm{d}t = F(s)$$

- Maps from t-space to s-space, where s is a complex-valued variable
- It is a linear transformation

Basic properties

Since it is a *linear transformation*:

1. $\mathscr{L}\{f+g\} = \mathscr{L}\{f\} + \mathscr{L}\{g\} = F(s) + G(s)$
2. $\mathscr{L}\{\beta f\} = \beta\mathscr{L}\{f\} = \beta F(s)$ where β is a constant

Exercise 3.3: Find the Laplace transform of $f(t) = 1$

$$\mathcal{L}\{1\} = \int_0^\infty \mathrm{e}^{-s\,t} \cdot 1 \; \mathrm{d}t = \lim_{a\to\infty} \int_0^a \mathrm{e}^{-s\,t} \; \mathrm{d}t$$

$$= \lim_{a\to\infty} \left[-\frac{1}{s} \mathrm{e}^{-s\,t} \right]_0^a = -\frac{1}{s} \lim_{a\to\infty} \left[\mathrm{e}^{-s\,a} - 1 \right] = \frac{1}{s}$$

Thus $\mathcal{L}\{1\} = F(s) = 1/s$

The Laplace transform exists (integral converges) if we restrict to $\mathrm{Re}(s) > 0$.

Exercise 3.4: Find the Laplace transform of $f(t) = \mathrm{e}^t$

$$\mathcal{L}\{\mathrm{e}^t\} = \int_0^\infty \mathrm{e}^{-s\,t}\,\mathrm{e}^t\;\mathrm{d}t = \lim_{a\to\infty}\int_0^a \mathrm{e}^{(1-s)\,t}\mathrm{d}t$$

$$= \lim_{a\to\infty}\left[\frac{\mathrm{e}^{(1-s)\,t}}{1-s}\right]_0^a = \frac{1}{s-1}$$

Thus $\mathcal{L}\{1\} = F(s) = 1/(s-1)$ for $\mathrm{Re}(s) > 1$.

Existence of Laplace transform / Convergence of improper integral

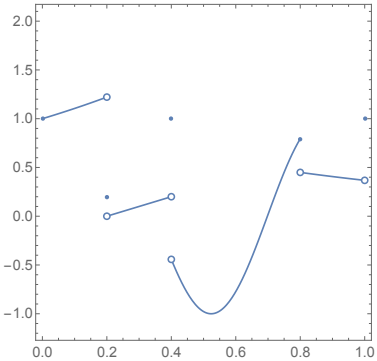
Criteria to ensure the Laplace transform exists (improper integral converges)

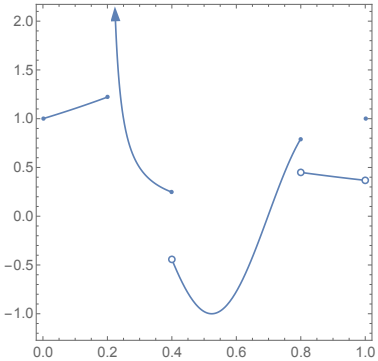
- f is piecewise continuous on $[0, L]$ for all $L > 0$
- $|f(t)| \leqslant k\,\mathrm{e}^{\alpha\,t}$ for all $t > T$ for $\alpha, k, T \in \mathbb{R}$
  - i.e., f bounded by an exponential function
  - Laplace transform guaranteed to converge for $\mathrm{Re}(s) > \alpha$

Piecewise continuous

- f has a finite number of discontinuities on $(0, L)$.
- The limits of f at the end points, and on each side of every discontinuity, are finite.

Exercise 3.5: Are the following functions piecewise continuous?





Exercise 3.6: Does the below function have a Laplace transform?

$$f = \begin{cases} 0 & 0 \leqslant t < 1 \\ 1 & 1 \leqslant t \end{cases}$$

- f contains one discontinuity (at $t = 1$)
- Limits at end points are bounded
- Therefore, it is piecewise continuous
- It is exponentially bounded for large t
- We deduce that f has a Laplace transform

Lecture 10 recap

In Lecture 10:

- Introduced improper integrals and learnt how to assess their convergence
- Defined the Laplace transform and discussed its linear properties
- Set out criteria for existence of the Laplace transform of a function

Lecture 11

In Lecture 11, we will:

1. Investigate existence of the Laplace transform for example functions
2. Introduce the inverse Laplace transform
3. Use the LT "cheat sheet" to calculate LT and ILT in examples

Exercise 3.6: Does the below function have a Laplace transform?

$$f = \begin{cases} 0 & 0 \leqslant t < 1 \\ 1 & 1 \leqslant t \end{cases}$$

- f contains one discontinuity (at $t = 1$)
- Limits at end points are bounded
- Therefore, it is piecewise continuous
- It is exponentially bounded for large t
- We deduce that f has a Laplace transform

Exercise 3.7: Do Laplace transforms exist for the following functions?

$$f_1(t) = t^3\,\mathrm{e}^{2t}; \qquad f_2(t) = \mathrm{e}^{t^2}$$

Let's examine if e^{3t} is greater than $f_1 = t^3\mathrm{e}^{2t}$ for large t.

$$\lim_{t\to\infty} \frac{t^3\mathrm{e}^{2t}}{\mathrm{e}^{3t}} = \lim_{t\to\infty} t^3\mathrm{e}^{-t} = 0$$

There will be some $T > 0$ at which e^{3t} becomes greater than $t^3\mathrm{e}^{2t}$.
Therefore, f_1 is bounded by an exponential and must have a Laplace transform.

f_2 grows faster than any exponential $k\,\mathrm{e}^{\alpha\,t}$ and so does not have a Laplace transform.

Inverse Laplace transform

Theorem

- Let f be a continuous function on $t \in [0, \infty)$ that has a Laplace transform, F
- Then the *inverse Laplace transform* also exists, i.e.,

$$\mathscr{L}^{-1}\{F\} = f \quad \text{for all } t \in [0, \infty)$$

The inverse Laplace transform is linear:

1. $\mathscr{L}^{-1}\{F + G\} = \mathscr{L}^{-1}\{F\} + \mathscr{L}^{-1}\{G\} = f + g$
2. $\mathscr{L}^{-1}\{\beta F\} = \beta \mathscr{L}^{-1}\{F\} = \beta f$ where β is a constant

*A formula for the inverse exists but it requires advanced complex analysis.

Discontinuous functions

1. If f is discontinuous at $t = 0$, the inverse returns the limit from above, i.e.,

$$\mathscr{L}^{-1}\{F\}\Big|_{t=0} = \lim_{t\to 0^+} f(t)$$

2. If f is discontinuous at $\tau > 0$, the inverse returns the mid-point

$$\mathscr{L}^{-1}\{F\}\Big|_{t=\tau} = \frac{1}{2}\left(\lim_{t\to \tau^-} f(t) + \lim_{t\to \tau^+} f(t)\right)$$

MAST20030 Differential Equations Formulae Sheet

1) Laplace Transforms

1.	$f(t)$	$F(s) = \mathscr{L}(f) = \int_0^\infty f(t)e^{-st}\,dt$ (Definition of Transform)
2.	1	$\frac{1}{s}$
3.	t^n	$\frac{n!}{s^{n+1}}$
4.	e^{at}	$\frac{1}{s-a}$
5.	$\sin(at)$	$\frac{a}{s^2+a^2}$
6.	$\cos(at)$	$\frac{s}{s^2+a^2}$
7.	$\sinh(at)$	$\frac{a}{s^2-a^2}$
8.	$\cosh(at)$	$\frac{s}{s^2-a^2}$

Exercise 3.8: Find the following Laplace transforms

$$\text{(i) } 2\mathrm{e}^{-3t} \qquad \text{(ii) } \sin(2t) + \cos(2t)$$

By linearity,

$$\begin{aligned}\mathscr{L}\{2\mathrm{e}^{-3t}\} &= 2\,\mathscr{L}\{\mathrm{e}^{-3t}\} \\ &= \frac{2}{s+3} \quad \text{from LT sheet}\end{aligned}$$

for $\mathrm{Re}(s) > -3$

Again, using linearity

$$\begin{aligned}\mathscr{L}\{\sin(2t) + \cos(2t)\} &= \mathscr{L}\{\sin(2t)\} + \mathscr{L}\{\cos(2t)\} \\ &= \frac{2}{s^2+4} + \frac{s}{s^2+4} = \frac{s+2}{s^2+4}\end{aligned}$$

for $\mathrm{Re}(s) > 0$

Exercise 3.9: Find the following inverse Laplace transforms

$$\text{(i)}\ \frac{2}{s^4} \qquad \text{(ii)}\ \frac{2}{s-1} + \frac{3s}{s^2+9}$$

$$\mathscr{L}^{-1}\left\{\frac{2}{s^4}\right\} = \frac{1}{3}\mathscr{L}^{-1}\left\{\frac{3!}{s^4}\right\} \quad \text{by linearity}$$

$$= \frac{t^3}{3} \quad \text{from LT sheet}$$

$$\mathscr{L}^{-1}\left\{\frac{2}{s-1} + \frac{3s}{s^2+9}\right\} = 2\,\mathscr{L}^{-1}\left\{\frac{1}{s-1}\right\} + 3\,\mathscr{L}^{-1}\left\{\frac{s}{s^2+3^2}\right\}$$

$$= 2\,\mathrm{e}^t + 3\,\cos(3t)$$

Partial fractions; **Exercise 3.10**: Find the inverse Laplace transform of

$$F(s) = \frac{2}{(s-1)(s-2)}$$

Seek A and B such that

$$F(s) = \frac{A}{s-1} + \frac{B}{s-2} = \frac{A(s-2)+B(s-1)}{(s-1)(s-2)}$$

$$\Rightarrow \quad 2 = A(s-2) + B(s-1) = (A+B)\,s - (2A+B)$$

$$\Rightarrow \quad A + B = 0 \quad \text{and} \quad 2A + B = -2$$

$$\Rightarrow \quad A = -B \quad \Rightarrow \quad 2A - A = A = -2 \quad \Rightarrow \quad B = 2$$

Thus

$$\mathscr{L}^{-1}\{F(s)\} = \mathscr{L}^{-1}\left\{\frac{-2}{s-1} + \frac{2}{s-2}\right\} = -2\,\mathscr{L}^{-1}\left\{\frac{1}{s-1}\right\} + 2\,\mathscr{L}^{-1}\left\{\frac{1}{s-2}\right\}$$

Lecture 11 recap

In Lecture 11:

- Learnt the existence of the inverse Laplace transform and its linear properties
- Learnt to calculate LTs and ILTs using the formulae "cheat" sheet + linearity
- Learnt to use partial fractions to calculate ILTs of suitable functions

Lecture 12

In Lecture 12, we will:

1. Learn how to express the LT of derivatives in a convenient form
2. Learn how to use the LT to solve IVPs
3. Learn the s-shifting and t-shifting theorems to calculate LTs and ILTs

Theorem (Laplace transform of the derivative)

Let f be a continuous function on $[0, \infty)$. If $\mathscr{L}\{f\}$ exists then

$$\mathscr{L}\{f'\} = s\,F(s) - f(0)$$

The initial condition is encoded into the Laplace transform!

Proof:

$$\begin{aligned}
\mathscr{L}\{f'\} &= \int_0^\infty \mathrm{e}^{-st} f'(t)\ \mathrm{d}t = \lim_{b\to\infty} \int_0^b \mathrm{e}^{-st} f'(t)\ \mathrm{d}t \\
&= \lim_{b\to\infty} \left\{ \Big[\mathrm{e}^{-st} f(t)\Big]_0^b - \int_0^b -s\,\mathrm{e}^{-st}\, f(t)\ \mathrm{d}t \right\} \\
&= -f(0) + s \lim_{b\to\infty} \int_0^b \mathrm{e}^{-st}\, f(t)\ \mathrm{d}t = s\,F(s) - f(0)
\end{aligned}$$

as required

Application to ODEs; **Exercise 3.11**: Solve $y'(t) - y(t) = 0$, $y(0) = 1$

Take Laplace transforms of both sides of the ODE:

$$\mathscr{L}\{y' - y\} = \mathscr{L}\{0\}$$

$$\Rightarrow \quad \underbrace{sY - y(0) - Y}_{(s-1)\,Y - 1} = 0$$

$$\Rightarrow \quad (s-1)\,Y = 1 \quad \Rightarrow \quad Y(s) = \frac{1}{s-1}$$

Take the inverse Laplace transform

$$\mathscr{L}^{-1}\{Y\} = \mathscr{L}^{-1}\left(\frac{1}{s-1}\right) = \mathrm{e}^{t} \quad \text{by LT table}$$

Theorem (Higher order derivatives)

Let $f, f', \ldots, f^{(n-1)}$ be continuous on $[0, \infty)$ and have Laplace transforms. Then

$$\mathscr{L}\left\{f^{(n)}\right\} = s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \ldots - f^{(n-1)}(0)$$

Can be proved using integration by parts and induction.

We will use the case of a second derivative extensively:

$$\mathscr{L}\{f''\} = s^2 F(s) - s f(0) - f'(0)$$

Exercise 3.12: $y''(t) - 3y'(t) + 2y(t) = 0$ where $y(0) = 0$ and $y'(0) = 2$

$$\mathscr{L}\{y'' - 3y' + 2y\} = \mathscr{L}\{0\}$$

$$\Rightarrow \quad \mathscr{L}\{y''\} - 3\,\mathscr{L}\{y'\} + 2\,\mathscr{L}\{y\} = 0$$

$$\Rightarrow \quad s^2\,Y - s\,y(0) - y'(0) - 3\left\{s\,Y - y(0)\right\} + 2\,Y = 0$$

$$\Rightarrow \quad \left(s^2 - 3\,s + 2\right)Y - 2 = 0$$

Therefore

$$Y = \frac{2}{s^2 - 3s + 2} = \frac{2}{(s-1)(s-2)}$$

$$\Rightarrow \quad y(t) = \mathscr{L}^{-1}\{Y(s)\} = 2(\mathrm{e}^{2t} - \mathrm{e}^{t}) \qquad \text{(Ex 3.10)}$$

Exercise 3.13: Find the Laplace transform of $\mathrm{e}^{3\,t}\,t^2$

Theorem (s-shift)

If $\mathscr{L}\{f\}$ exists for $\mathrm{Re}(s) > \alpha$, then

$$\mathscr{L}\{\mathrm{e}^{\omega\,t} f(t)\} = F(s-\omega), \qquad \mathrm{Re}(s) > \alpha + \omega$$

$$\mathscr{L}\{t^2\} = \frac{2}{s^3} \quad \text{for} \quad \mathrm{Re}(s) > 0$$

$$\Rightarrow \quad \mathscr{L}\{\mathrm{e}^{3t}\,t^2\} = \frac{2}{(s-3)^3} \quad \text{for} \quad \mathrm{Re}(s) > 3$$

Exercise 3.14: Find the inverse Laplace transform of $\dfrac{3}{(s+2)^2+9}$

$$\mathscr{L}^{-1}\left(\frac{3}{s^2+9}\right) = \sin 3t \qquad \text{from LT table}$$

$$\Rightarrow \quad \mathscr{L}^{-1}\left(\frac{3}{(s+2)^2+9}\right) = \mathrm{e}^{-2t}\sin 3t \qquad \text{by } s\text{-shifting}$$

Where is the shifted transform valid?

Discontinuous functions

Definition (Heaviside step function)

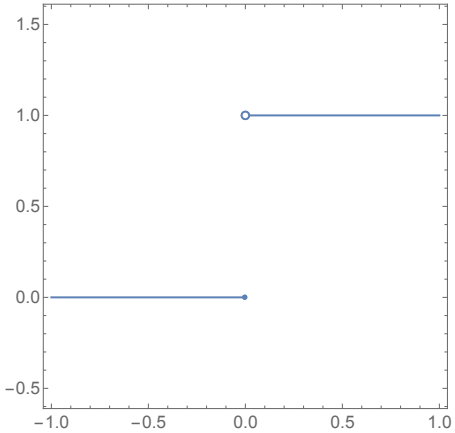
$$H(t) = \begin{cases} 0 & t \leqslant 0 \\ 1 & t > 0 \end{cases}$$

The value of $H(0)$ is usually irrelevant.

It can be translated so the discontinuity occurs at $t = c$, i.e.,

$$H(t - c)$$

It can be used to model turning processes on or off.



Exercise 3.15: Find the Laplace transform of $f = \sin(t-3)H(t-3)$

Theorem (t-shift)

If $F(s)$ exists for $\mathrm{Re}(s) > \alpha$, and $c \geqslant 0$, then

$$\mathscr{L}\Big\{f(t-c)H(t-c)\Big\} = \mathrm{e}^{-cs}F(s), \quad \mathrm{Re}(s) > \alpha$$

$$F(s) = \mathrm{e}^{-3s}\mathscr{L}(\sin t)$$

$$\Rightarrow \quad F(s) = \frac{\mathrm{e}^{-3s}}{s^2+1} \quad \text{for} \quad \mathrm{Re}(s) > 0$$

Exercise 3.16: Find the ILT of $\dfrac{\mathrm{e}^{-2s}}{s-4}$ where $\mathrm{Re}(s) > 4$

$$\mathscr{L}^{-1}\left\{\frac{1}{s-4}\right\} = \mathrm{e}^{4t} \quad \text{for} \quad \mathrm{Re}(s) > 4$$

$$\Rightarrow \quad \mathscr{L}^{-1}\left\{\frac{\mathrm{e}^{-2s}}{s-4}\right\} = \mathrm{e}^{4(t-2)}H(t-2) \quad \text{for} \quad \mathrm{Re}(s) > 4$$

Lecture 12 recap

In Lecture 12:

- Learnt how the LT maps derivative in t-space to multiplication in s-space
- Understood how the LT encodes initial values
- Learnt how to solve IVPs using the LT and ILT
- Introduced the Heaviside step function
- Learnt how to use the s- and t-shifting theorems to calculate LTs and ILTs

Lecture 13

In Lecture 13, we will:

1. Learn how to solve IVPs involving a forcing that "switches on"
2. Introduce the Dirac delta

Discontinuous IVPs; **Exercise 3.17**: $y'(t) + y(t) = H(t-2)$, $y(0) = 1$

$H(t-2)$ on right-hand side $\Rightarrow$ forcing function switches on at $t = 2$.

$$\underbrace{\mathscr{L}\{y'(t) + y(t)\}}_{sY - 1 + Y} = \underbrace{\mathscr{L}\{H(t-2)\}}_{\mathrm{e}^{-2s}\mathscr{L}(1)} = \frac{\mathrm{e}^{-2s}}{s} \qquad t\text{-shifting}$$

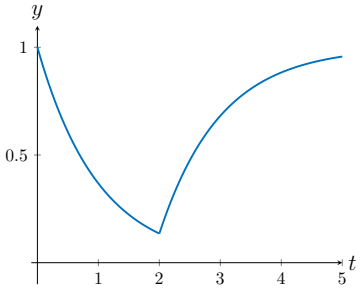
$$\Rightarrow \quad Y = \frac{1}{s+1} + \frac{\mathrm{e}^{-2s}}{s(s+1)}$$

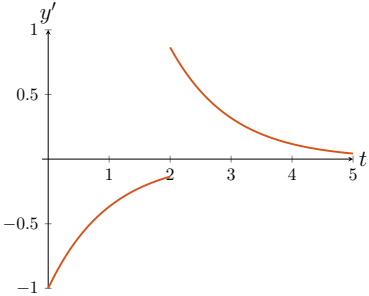
$$\frac{1}{s(s+1)} = \frac{A}{s} + \frac{B}{s+1} \;\Rightarrow\; A(s+1) + Bs = 1 \;\Rightarrow\; A = 1, B = -1$$

$$\Rightarrow \quad Y = \frac{1}{s+1} + \frac{\mathrm{e}^{-2s}}{s} - \frac{\mathrm{e}^{-2s}}{s+1}$$

$$y = \underbrace{\mathscr{L}^{-1}\left\{\frac{1}{s+1}\right\}}_{\mathrm{e}^{-t}} + \underbrace{\mathscr{L}^{-1}\left\{\frac{\mathrm{e}^{-2s}}{s}\right\}}_{H(t-2)} - \underbrace{\mathscr{L}^{-1}\left\{\frac{\mathrm{e}^{-2s}}{s+1}\right\}}_{\mathrm{e}^{-(t-2)}H(t-2)} = \mathrm{e}^{-t} + (1-\mathrm{e}^{2-t})H(t-2)$$

$$\lim_{t\to 2^-} y(t) = \lim_{t\to 2^+} y(t) = \mathrm{e}^{-2} \quad \text{but} \quad \lim_{t\to 2^-} y'(t) = -\mathrm{e}^{-2} \neq 1-\mathrm{e}^{-2} = \lim_{t\to 2^+} y'(t)$$





Solution to IVPs with discontinuous forcing

Consider the nth-order ODE

$$y^{(n)}(t) + f_{n-1}(t)y^{(n-1)} + \ldots + f_0(t)y(t) = g(t)$$

- Let $f_0, \ldots, f_n$ be continuous on $[0, \infty)$
- Let g be piecewise continuous on $[0, L]$ for all $L > 0$

ODE has solutions, such that:

1. $y, y', \ldots, y^{(n-1)}$ are continuous on $[0, \infty)$
2. $y^{(n)}$ is continuous everywhere g is
3. The left and right limits of $y^{(n)}$ are defined by the ODE at all discontinuities
4. y satisfies the ODE at all points of continuity of g

Exercise 3.18: Solve the below IVP

$$y'' + 4y = g(t) \quad \text{where} \quad g(t) = \begin{cases} t & 0 < t \leqslant 2 \\ 0 & t > 2 \end{cases} \quad \text{and} \quad y(0) = y'(0) = 0$$

Rewrite the discontinuous term

$$g(t) = t - t\,H(t-2) = t - 2\,H(t-2) - (t-2)\,H(t-2)$$

Taking the Laplace transform of both sides:

$$\mathscr{L}\{y'' + 4\,y\} = s^2 Y + 4Y = \underbrace{\mathscr{L}\{t\}}_{\frac{1}{s^2}} - 2\,\underbrace{\mathscr{L}\{H(t-2)\}}_{\frac{\mathrm{e}^{-2s}}{s}} - \underbrace{\mathscr{L}\{(t-2)\,H(t-2)\}}_{\frac{\mathrm{e}^{-2s}}{s^2}}$$

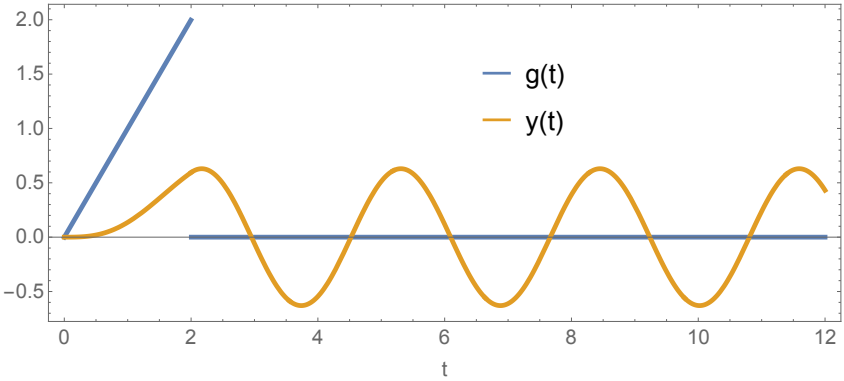
$$\Rightarrow \quad Y(s) = \frac{1}{s^2(s^2+4)} - \frac{2\,s+1}{s^2\,(s^2+4)}\,\mathrm{e}^{-2s}$$

Use partial fractions to write Y in a more convenient form

$$Y = \frac{1}{4s^2} - \frac{1}{4(s^2+4)} - \mathrm{e}^{-2s}\left(\frac{1}{2s} + \frac{1}{4s^2} - \frac{s}{2(s^2+4)} - \frac{1}{4(s^2+4)}\right)$$

$$\begin{aligned}
\Rightarrow \quad y(t) = \mathscr{L}^{-1}\{Y(s)\} &= \frac{1}{4}\mathscr{L}^{-1}\left\{\frac{1}{s^2}\right\} - \frac{1}{8}\mathscr{L}^{-1}\left\{\frac{2}{s^2+2^2}\right\} - \frac{1}{2}\mathscr{L}^{-1}\left\{\frac{\mathrm{e}^{-2s}}{s}\right\} \\
&\quad - \frac{1}{4}\mathscr{L}^{-1}\left\{\frac{\mathrm{e}^{-2s}}{s^2}\right\} + \frac{1}{2}\mathscr{L}^{-1}\left\{\frac{s\,\mathrm{e}^{-2s}}{s^2+4}\right\} + \frac{1}{8}\mathscr{L}^{-1}\left\{\frac{2\,\mathrm{e}^{-2s}}{s^2+4}\right\} \\
&= \frac{1}{4}t - \frac{1}{8}\sin 2t \\
&\quad - \left(\frac{1}{2} + \frac{t-2}{4} - \frac{1}{2}\cos 2(t-2) - \frac{1}{8}\sin 2(t-2)\right) H(t-2)
\end{aligned}$$

$$y(t) = \frac{1}{4}t - \frac{1}{8}\sin 2t - \left(\frac{1}{2} + \frac{t-2}{4} - \frac{1}{2}\cos 2(t-2) - \frac{1}{8}\sin 2(t-2)\right) H(t-2)$$

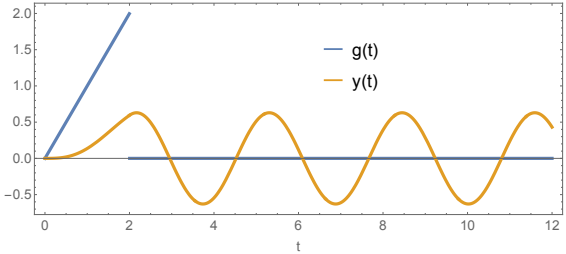


Lecture 13 recap

In Lecture 13:

- Went through examples on using LTs to solve IVPs with discontinuous forcing
  - Looked at off–on forcing and how this produces discontinuities in a solution derivative
  - Learnt how to represent a ramp-like forcing using Heaviside step functions

$$y(t) = \frac{1}{4}t - \frac{1}{8}\sin 2t - \left(\frac{t}{4} - \frac{1}{2}\cos 2(t-2) - \frac{1}{8}\sin 2(t-2)\right) H(t-2)$$



Lecture 14

In Lecture 14, we will:

1. Introduce the Dirac delta and use it to model impulsive forcing
2. Introduce convolutions as a tool to calculate ILTs

Impulses

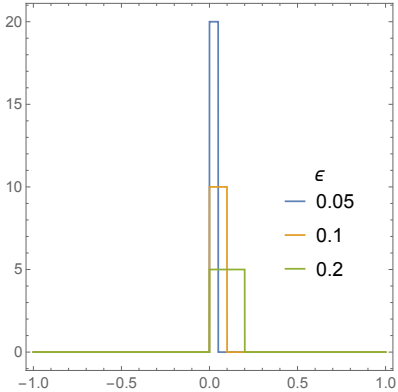
Suppose we want to model a force acting over a negligibly small time period, i.e., an impulse

Use a rectangular pulse of unit area:

$$\delta_\epsilon(t) = \begin{cases} 0 & t \leqslant 0 \\ \frac{1}{\epsilon} & 0 < t \leqslant \epsilon \\ 0 & t > \epsilon \end{cases}, \qquad \text{“small” } \epsilon > 0$$

The ‘strength’ of this impulse is

$$\int_0^\infty \delta_\epsilon(t)\,\mathrm{d}t = \int_0^\epsilon \frac{1}{\epsilon}\,\mathrm{d}t = 1$$



As width $\to 0$, height $\to \infty$ but area remains 1

Limit as $\epsilon \to 0$

1. $\lim_{\epsilon \to 0} \delta_\epsilon = 0$ for all $t \neq 0$
2. $\lim_{\epsilon \to 0} \int_0^\infty \delta_\epsilon \, \mathrm{d}t = 1$
3. $\lim_{\epsilon \to 0} \mathscr{L}(\delta_\epsilon) = \lim_{\epsilon \to 0} \dfrac{1 - \mathrm{e}^{-\epsilon s}}{\epsilon s} = 1$ (L'Hôpital's rule)

No function (in the traditional sense) can have these properties!

The Dirac delta, $\delta(t)$

Definition

We define $\delta(t)$ such that

1. $\delta(t) = 0$ for all $t \neq 0$,
2. $\displaystyle\int_{-\infty}^{\infty} \delta(t)\mathrm{d}t = \int_{0}^{a} \delta(t)\mathrm{d}t = 1$ for all $a > 0$, and
3. $\mathscr{L}\{\delta(t)\} = 1$.

- $\delta(t)$ is not a function in the traditional sense (it is a *generalised function* as you will discover in later courses)
- There are various forms of $\delta(t)$ with others being centred at or left of zero; we will always use the above definition

Sifting property

- Since $\delta(t)$ is zero for $t \neq 0$, it 'sifts out' the value of a function at $t = 0$

$$\int_0^\infty f(t)\delta(t)\mathrm{d}t = f(0)$$

- Let $c \geqslant 0$, then

1. $\displaystyle\int_0^\infty f(t)\delta(t-c)\ \mathrm{d}t = f(c)$

2. $\displaystyle\mathscr{L}\Big\{\delta(t-c)\Big\} = \int_0^\infty \mathrm{e}^{-st}\delta(t-c)\ \mathrm{d}t = \mathrm{e}^{-sc}$

Exercise 3.19: Solve $y'(t) - y(t) = \delta(t-1)$ where $y(0) = 0$

We solve the problem using Laplace transforms.

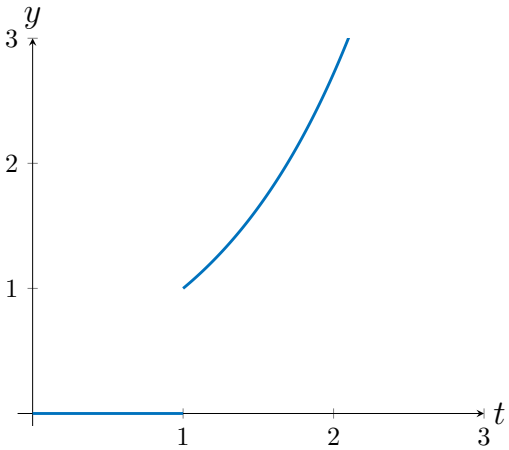
$$\mathscr{L}\{y' - y\} = \mathscr{L}\{\delta(t-1)\}$$

$$\Rightarrow \quad sY - Y = \mathrm{e}^{-s}$$

$$Y(s) = \frac{\mathrm{e}^{-s}}{s-1}$$

$$y(t) = \mathscr{L}^{-1}\left(\frac{\mathrm{e}^{-s}}{s-1}\right) = \mathrm{e}^{t-1}H(t-1) \quad t\text{-shifting}$$

$$y(t) = \mathrm{e}^{t-1} H(t-1)$$

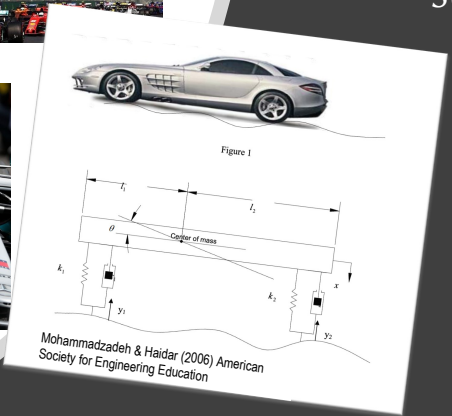


The solution satisfies the initial condition and the ODE but it is discontinuous

Case study 4: Formula 1 car suspension

https://www.planetf1.com/news/sydney-australian-grand-prix-bid/

https://www.mercedesamgf1.com/en/news/2019/06/the-suspension-of-a-formula-one-car/



- Read all about formula 1 car suspensions here https://www.mercedesamgf1.com/en/news/2019/06/the-suspension-of-a-formula-one-car/
- **Mathematical models lead to ODEs which can be solved using Laplace transforms.**

Convolution

Definition (Convolution)

Given two functions, f and g, defined on $[0, \infty)$, their convolution, $f * g$ is defined

$$f * g = \int_0^t f(t-\tau)g(\tau)\mathrm{d}\tau$$

Basic properties

- Convolutions are commutative, i.e., $f * g = g * f$
- In general, $1 * f \neq f$, as

$$(1 * f)(t) = \int_0^t f(\tau)\mathrm{d}\tau, \quad \text{i.e., an antiderivative}$$

Exercise 3.20: Calculate the convolution of e^t with e^{2t}

$$\begin{aligned}
\mathrm{e}^t * \mathrm{e}^{2t} &= \int_0^t \mathrm{e}^{t-\tau}\mathrm{e}^{2\tau}\ \mathrm{d}\tau \\
&= \mathrm{e}^t \int_0^t \mathrm{e}^{\tau}\mathrm{d}\tau \\
&= \mathrm{e}^t(\mathrm{e}^t - 1)
\end{aligned}$$

Note that

$$\mathrm{e}^{2t} * \mathrm{e}^t = \int_0^t \mathrm{e}^{2(t-\tau)}\mathrm{e}^{\tau}\ \mathrm{d}\tau = \mathrm{e}^{2t}\int_0^t \mathrm{e}^{-\tau}\ \mathrm{d}\tau = \mathrm{e}^{2t}(1-\mathrm{e}^{-t}) = \mathrm{e}^t(\mathrm{e}^t - 1)$$

Exercise 3.21: Calculate $\mathscr{L}^{-1}\left\{\dfrac{1}{(s-1)(s-2)}\right\}$

Laplace transform of the convolution

Let f and g be functions with Laplace transforms, then

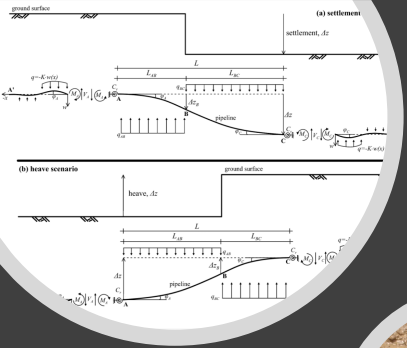
$$\mathscr{L}(f * g) = F(s)G(s)$$

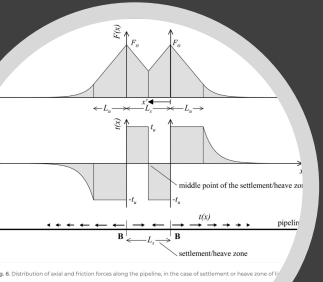
Laplace transforms convert convolutions into multiplication

$$\mathscr{L}^{-1}\left\{\frac{1}{s-1}\right\} = \mathrm{e}^{t} \quad \text{and} \quad \mathscr{L}^{-1}\left\{\frac{1}{s-2}\right\} = \mathrm{e}^{2t}$$

$$\Rightarrow \quad \mathscr{L}^{-1}\left\{\frac{1}{s-1}\frac{1}{s-2}\right\} = \mathrm{e}^{t} * \mathrm{e}^{2t}$$

From previous exercise $\mathrm{e}^{t} * \mathrm{e}^{2t} = \mathrm{e}^{t}(\mathrm{e}^{t} - 1)$





g. 6. Distribution of axial and friction forces along the pipeline, in the case of settlement or heave zone of l

Case study 5: Critical pipeline performance

- Buried pipelines are critical lifeline systems, supplying drinking water and natural gas to communities. Damage to these pipelines caused by ground deformation and related hazards like earthquakes and landslides can lead to severe socioeconomic losses to unprepared communities.
- Mitigation measures employ mathematical models of these structures and these often involve ordinary differential equations with discontinuous forcing and impulsive functions
- Images from Lee et al (2016) Quarterly Journal of Engineering Geology and Hydrogeology, and Kouretzis et al (2015) Canadian Geotechnical Journal

Exercise 3.22: Solve the following integro-differential equation

$$y'(t) + 4\int_0^t y(\tau)\mathrm{d}\tau = 0 \qquad \text{where} \qquad y(0) = 3$$

Laplace transform of an antiderivative

$$\mathscr{L}\left(\int_0^t f(\tau)\mathrm{d}\tau\right) = \mathscr{L}(1 * f) = \mathscr{L}(1)F(s) = \frac{F(s)}{s}$$

Taking the Laplace transform gives

$$\begin{aligned}
sY - 3 + 4Y * \mathscr{L}(1) = 0 \quad &\Rightarrow \quad sY - 3 + \frac{4Y}{s} = 0 \\
s^2Y - 3s + 4Y = 0 \quad &\Rightarrow \quad (s^2+4)Y = 3s \\
Y = \frac{3s}{s^2+4} \quad &\Rightarrow \quad Y = \mathscr{L}^{-1}\left(\frac{3s}{s^2+4}\right) = 3\cos 2t
\end{aligned}$$